

由定义,有:

$$\forall \varepsilon > 0, \exists N \in \mathbb{Z} \quad |a_n - l| < \varepsilon$$

$$|a_n - l| \leq |a_n - l| < \varepsilon$$

则 $\forall \varepsilon > 0$, 当 $n > N$ 时

$$||a_n| - |l|| < \varepsilon$$

$$\therefore \lim_{n \rightarrow \infty} |a_n| = |l|$$

3. 设 $\{a_n\}$ 有极限 l , 证明

(1) 存在一个自然数 N , $n < N$ $|a_n| \leq |l| + 1$;

(2) $\{a_n\}$ 是一个有界数列, 即存在一个常数 M , 使得 $|a_n| \leq M$ ($n = 1, 2, \dots$).

修正: $n > N$ $|a_n| \leq |l| + 1$

(1) 由定义, 有: $\forall \varepsilon \in (0, 1) \exists n_0 \in \mathbb{Z}$, 当 $n > n_0$ 时, $|a_n - l| < \varepsilon$

$$||a_n| - |l|| \leq |a_n - l| < \varepsilon$$

$$||a_n| - |l|| < \varepsilon$$

$$-\varepsilon < |a_n| - |l| < \varepsilon < 1$$

$$\therefore |a_n| \leq |l| + 1$$

或 当 $n < n_0$ 时

$$\text{取 } M_0 = \max_{i=1}^n |a_i|$$

当 $n > n_0$ 时

$$|a_n| \leq |l| + 1$$

则取 $M = \max(|l| + 1, M_0)$ 即可

$$(4) \lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0;$$

$$\frac{1}{n_0} < \varepsilon$$

$$n_0 > \frac{1}{\varepsilon}$$

$$\text{取 } n_0 = \lceil \frac{1}{\varepsilon} \rceil \text{ 即可}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$$

$$\text{又 } \frac{n!}{n^n} \geq 0$$

$$\therefore \lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$$

$$\frac{n!}{n^n} = \frac{n}{n} \cdot \frac{n-1}{n} \cdots \frac{1}{n}$$

$$\leq \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \frac{n!}{n^n} \leq \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{n!}{n^n} \leq \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \quad \therefore \lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$$

证明 $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$

当 $n > n_0$ 时

$$\forall \varepsilon \in (0, 1) \exists n_0 \text{ s.t. } \left| \frac{1}{n} - 0 \right| < \varepsilon$$

$$(5) \lim_{n \rightarrow \infty} \left(\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{(n-1) \cdot n} \right) = 1;$$

$$\sum_{i=1}^{n-1} \frac{1}{i(i+1)}$$

$$= \sum_{i=1}^{n-1} \frac{1}{i} - \frac{1}{i+1}$$

$$= 1 - \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \text{原式}$$

$$= \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n} \right)$$

$$= 1 - \lim_{n \rightarrow \infty} \frac{1}{n}$$

$$= 1 - 0 = 1$$

$$(6) \lim_{n \rightarrow \infty} \left(\frac{1}{(n+1)^{3/2}} + \cdots + \frac{1}{(2n)^{3/2}} \right) = 0.$$

类似提示:

$$\sum_{i=1}^n \frac{1}{(n+i)^{3/2}}$$

$$\leq n \cdot \frac{1}{n^{3/2}}$$

$$= \frac{1}{\sqrt{n}}$$

$$\lim_{n \rightarrow \infty} \text{原式} \leq \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0$$

$$\text{证明 } \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0$$

$$\text{任意设 } \varepsilon \in (0, 1)$$

$$\frac{1}{\sqrt{n_0}} < \varepsilon$$

$$\frac{1}{n_0} < \varepsilon^2$$

$$n_0 > \frac{1}{\varepsilon^2}$$

$$\text{取 } n_0 = \left\lceil \frac{1}{\varepsilon^2} \right\rceil$$

$$\text{则当 } n > n_0 \text{ 时, } \frac{1}{\sqrt{n}} < \varepsilon$$

$$\text{又: 原式} > 0$$

$$\therefore 0 < \text{原式} < \frac{1}{\sqrt{n}}$$

用夹极限

$$\lim_{n \rightarrow \infty} 0 < \lim_{n \rightarrow \infty} \text{原式} < \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}}$$

由夹逼定理得

$$\lim_{n \rightarrow \infty} \text{原式} = 0$$

$$(3) \lim_{n \rightarrow \infty} n^2 q^n = 0 (|q| < 1);$$

$$\text{设 } a_n = n^2 q^n$$

当 $n > n_0$ 时

a_n 单调

$$\therefore |q| - 1 \neq 0$$

$$\therefore x = 0$$

$$\text{设 } a_n = n^2 q^n$$

当 $n > n_0$ 时

a_n 单调

又 $a_n > 0$

$\therefore a_n$ 为收敛数列

$$a_{n+1} = \left(\frac{n+1}{n}\right)^2 \cdot |q| \cdot a_n$$

同取极限: 设 $\lim_{n \rightarrow \infty} a_n = X$

$$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n}\right)^2 \cdot \lim_{n \rightarrow \infty} (a_n) \cdot |q|$$

$$X = X \cdot |q|$$

$$X(|q| - 1) = 0$$

$$\therefore |q| - 1 \neq 0$$

$$\therefore X = 0$$

$$\therefore \lim_{n \rightarrow \infty} a_n = 0$$

$$1^\circ \text{ 若 } q > 0, \text{ 则 } n^2 q^n = n^2 \cdot |q|^n \checkmark$$

$$\text{若 } q < 0, \text{ 则 } 2|n| \text{ 时 } n^2 q^n = n^2 |q|^n \checkmark$$

若 $q < 0$ 且 $2 \nmid n$

$$n^2 q^n = -n^2 |q|^n, \text{ 同取极限}$$

$$\lim_{n \rightarrow \infty} n^2 q^n = \lim_{n \rightarrow \infty} (-n^2 |q|^n)$$

$$\lim_{n \rightarrow \infty} n^2 q^n = 0 \text{ 得证}$$

$$\frac{a_{n+1}}{a_n} = \left(\frac{n+1}{n}\right)^2 \cdot |q|$$

$$\left(\frac{n+1}{n}\right)^2 \cdot |q| < 1$$

$$\Leftrightarrow \frac{n+1}{n} < \sqrt{\frac{1}{|q|}}$$

$$\Leftrightarrow \frac{1}{n} < \sqrt{\frac{1}{|q|}} - 1$$

$$\Leftrightarrow n > \frac{1}{\sqrt{\frac{1}{|q|}} - 1}$$

$$\text{设 } n_0 = \left\lceil \frac{1}{\sqrt{\frac{1}{|q|}} - 1} \right\rceil + 1$$

5. 设 $\lim_{n \rightarrow \infty} a_n = 0$, $\{b_n\}$ 是有界数列, 即存在常数 M , 使得 $|b_n| < M (n = 1, 2, \dots)$, 证明 $\lim_{n \rightarrow \infty} a_n b_n = 0$.

即证 $\forall \varepsilon > 0, \exists N, \text{ s.t. 当 } n > N \text{ 时 } |a_n \cdot b_n - 0| < \varepsilon$

任取 $\varepsilon > 0$, 取 $\varepsilon_1 = \frac{\varepsilon}{M}$

则由 $\lim_{n \rightarrow \infty} a_n = 0$ 得存在 N_0 满足当 $n > N_0$ 时 $|a_n| < \varepsilon_1$

则当 $n > N_0$ 时

$$|a_n \cdot b_n| = |a_n| \cdot |b_n|$$

$$< \frac{\varepsilon}{M} \cdot M$$

$$= \varepsilon$$

满足条件.

得证

6. 证明 $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$.

$$= \left(e^{\frac{1}{n}} \right)^{\frac{1}{n}}$$

$$= e^{\left(\frac{\ln n}{n}\right)}$$

$$\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = e^{\lim_{n \rightarrow \infty} \left(\frac{\ln n}{n}\right)}$$

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n} = \lim_{n \rightarrow \infty} \frac{(\ln n)'}{(n)'}$$

$$= \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{1}$$

$$= 0$$

$$\therefore \lim_{n \rightarrow \infty} n^{\frac{1}{n}} = e^0 = 1$$

$$= 0$$

$$(1) \lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}).$$

$$= \frac{\sqrt{n+1} - \sqrt{n}}{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}$$

$$= \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

$$= 0$$

$$(2) \lim_{n \rightarrow \infty} \frac{n^3 + 3n^2 - 100}{4n^3 - n + 2}.$$

$$= \lim_{n \rightarrow \infty} \frac{1 + 3(\frac{1}{n}) - 100(\frac{1}{n^3})}{4 - (\frac{1}{n^2}) + 2(\frac{1}{n^3})}$$

$$= \frac{1 + 3 \lim_{n \rightarrow \infty} \frac{1}{n} - 100 \lim_{n \rightarrow \infty} \frac{1}{n^3}}{4 - \lim_{n \rightarrow \infty} \frac{1}{n^2} + 2 \lim_{n \rightarrow \infty} \frac{1}{n^3}}$$

$$= \frac{1 + 0 - 0}{4 - 0 + 0}$$

$$= \frac{1}{4}$$

$$= \frac{1}{4}$$

$$(3) \lim_{n \rightarrow \infty} \frac{(2n+10)^4}{n^4 + n^3}.$$

$$= \lim_{n \rightarrow \infty} \frac{(2 + \frac{10}{n})^4}{1 + \frac{1}{n}}$$

$$= \frac{\lim_{n \rightarrow \infty} (2 + \frac{10}{n})^4}{1 + \lim_{n \rightarrow \infty} \frac{1}{n}}$$

$$= 16$$

$$(4) \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^{-2n}$$

$$\left(1 + \frac{1}{n}\right)^{-2n} = \frac{1}{\left(1 + \frac{1}{n}\right)^{2n}}$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

$$\therefore \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n}\right)^{2n}} = \frac{1}{e^2}$$

8. 利用单调有界序列有极限证明下列序列极限的存在性:

$$(2) x_n = \frac{1}{2+1} + \frac{1}{2^2+1} + \cdots + \frac{1}{2^n+1} \quad (n=1, 2, \dots);$$

$$(4) x_n = 1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{n!} \quad (n=1, 2, \dots).$$

$$(2) x_n = \sum_{i=1}^n \frac{1}{2^i+1}$$

$$< \sum_{i=1}^n \frac{1}{2^i}$$

$$= 1 - \frac{1}{2^n}$$

$$< 1$$

$$x_{n+1} - x_n = \frac{1}{2^{n+1}+1} > 0$$

x_n 单调递增且 $x_n < 1, n=1, 2, \dots$

x_n 存在极限

$$(4) x_n = \sum_{i=0}^n \frac{1}{i!}$$

$$< 1 + 1 + \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \cdots + \frac{1}{(n-1)n}$$

$$= 2 + 1 - \frac{1}{n}$$

$$< 3$$

$$x_{n+1} - x_n = \frac{1}{(n+1)!} > 0$$

x_n 单调且 $x_n < 3, n=1, 2, \dots$

x_n 存在极限